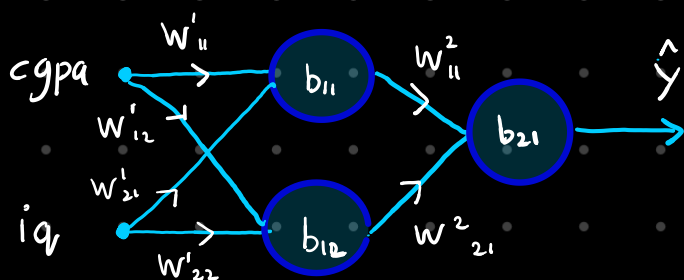


Lecture 4: Neural Networks and Backpropagation

Backpropagation is an algorithm for supervised learning of artificial neural network using gradient descent.



$$O_{11} = (w'_{11} \cdot \text{cgpa} + w'_{21} \cdot \text{iq}) + b_{11}$$

$$O_{12} = (w'_{12} \cdot \text{cgpa} + w'_{22} \cdot \text{iq}) + b_{12}$$

$$\Rightarrow O_{21} = (w''_{11} \cdot O_{11} + w''_{21} \cdot O_{12}) + b_{21}$$

Steps:

- ① select a data point (row)
- ② Predict using current weights (forward prop)
- ③ Choose a loss-function and compute loss. (MSE)
- ④ Update weights and biases (Gradient Decent)

$$W_{\text{new}} = W_{\text{old}} - \eta \frac{\partial L}{\partial W_{\text{old}}}$$

E.g. $\rightarrow W''_{11} \text{ new} = W''_{11} \text{ old} - \eta \frac{\partial L}{\partial W''_{11}}$

"derivate of loss w.r.t the neural network's weights."

$$\frac{\partial L}{\partial W''_{11}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial W''_{11}}$$

$$\Rightarrow -2(Y - \hat{y}) \cdot O_{11}$$

$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} (Y - \hat{y})^2 = -2(Y - \hat{y})$$

$$\frac{\partial \hat{y}}{\partial W''_{11}} = \frac{\partial O_{21}}{\partial W''_{11}} = \frac{\partial [O_{11} W''_{11} + O_{12} W''_{21} + b_{21}]}{\partial W''_{11}} = O_{11}$$

$$\frac{\partial L}{\partial W''_{21}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial W''_{21}}$$

$$\Rightarrow -2(Y - \hat{y}) \cdot O_{12}$$

$$\frac{\partial \hat{y}}{\partial W''_{21}} = \frac{\partial [O_{11} W''_{11} + O_{12} W''_{21} + b_{21}]}{\partial W''_{21}} = O_{12}$$

$$\frac{\partial L}{\partial W'_{11}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_{11}} \times \frac{\partial O_{11}}{\partial W'_{11}}$$

$$\frac{\partial \hat{y}}{\partial O_{11}} = \frac{\partial [O_{11} W''_{11} + O_{12} W''_{21} + b_{21}]}{\partial O_{11}} = W''_{11}$$

$$\frac{\partial O_{11}}{\partial W'_{11}} = \frac{\partial [cgpa \cdot W'_{11} + w'_{21} \cdot iq + b_{11}]}{\partial W'_{11}} = cgpa = x_{11}$$



Regression Problem

cgpa	resume score	package (lpa)
8	8	4
7	9	5
6	10	6
5	12	7

$$\textcircled{1} \frac{\partial L}{\partial W_{11}^2} = -2(y - \hat{y}) \cdot O_{11}$$

$$\textcircled{2} \frac{\partial L}{\partial W_{21}^2} = -2(y - \hat{y}) \cdot O_{12}$$

$$\textcircled{3} \frac{\partial L}{\partial b_{21}} = -2(y - \hat{y})$$

Backpropagation Algorithm

for i in range (epochs):

for j in range (X.shape[0]):

→ select jth row

→ Predict (forward prop)

→ calculate loss

→ Update weights and bias using GD

$$W_{\eta} = W_0 - \eta \frac{\partial L}{\partial W}$$

$$\textcircled{4} \frac{\partial L}{\partial W_{11}^2} = -2(y - \hat{y}) W_{11}^2 \cdot x_{i1}$$

$$\textcircled{5} \frac{\partial L}{\partial W_{21}^2} = -2(y - \hat{y}) W_{21}^2 \cdot x_{i2}$$

$$\textcircled{6} \frac{\partial L}{\partial b_{11}} = -2(y - \hat{y}) W_{11}^2$$

$$\textcircled{7} \frac{\partial L}{\partial W_{12}^2} = -2(y - \hat{y}) W_{12}^2 \cdot x_{i1}$$

$$\textcircled{8} \frac{\partial L}{\partial W_{22}^2} = -2(y - \hat{y}) W_{22}^2 \cdot x_{i2}$$

$$\textcircled{9} \frac{\partial L}{\partial b_{12}} = -2(y - \hat{y}) W_{12}^2$$

→ calculate average loss for the epoch

→ Loss function is a function of all trainable params.

$$\rightarrow L = (y - \hat{y})^2 \Rightarrow L(\hat{y})$$

$$\hat{y} = W_{11}^2 O_{11} + W_{21}^2 O_{12} + b_{21}$$

$$\hat{y} = W_{11}^2 (W_{11}' x_{i1} + W_{21}' x_{i2} + b_{11}) + W_{21}^2 (W_{12}' x_{i1} + W_{22}' x_{i2} + b_{12}) + b_{21}$$

Concept of gradient

$$y = b(x) = x^2 + x$$

$$\frac{dy}{dx} = 2x + 1$$

$$Z = b(x, y) = x^2 + y^2$$

$$\left. \begin{aligned} \frac{\partial Z}{\partial x} &= 2x \\ \frac{\partial Z}{\partial y} &= 2y \end{aligned} \right\} \text{gradient}$$

